1. Cell Structure

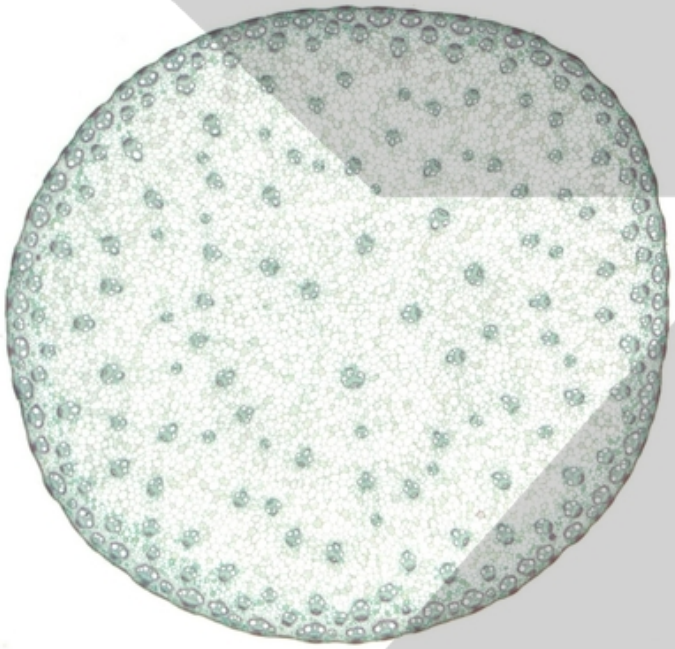
1.1. The Microscope in Cell Studies

Making temporary slides:

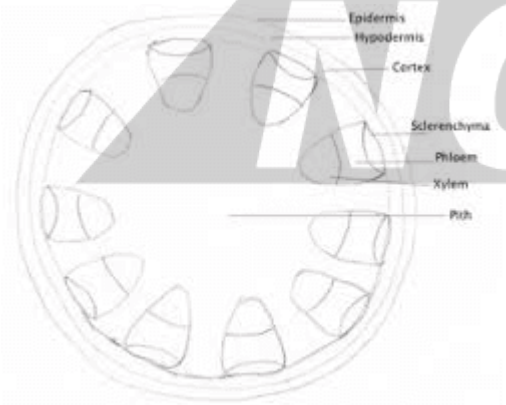
- Cut thin sections of the material to be examined
- Stain
- Mount the specimen on a clean glass slide
- Lower the coverslip over the specimen (to prevent the specimen drying out)

Photomicrograph vs Drawing:

Photomicrograph of a stem



Plan drawing of a stem



Types of Microscopes:

Feature	Light	Electron
Source of radiation	Light	Electrons
Wavelength of radiation	400-700nm	0.1nm
Maximum resolution	200nm	at least 0.05nm
Specimen	Alive	Dead
Image	Coloured photomicrograph	Black and white electron micrograph

Two types of Electron Microscope:

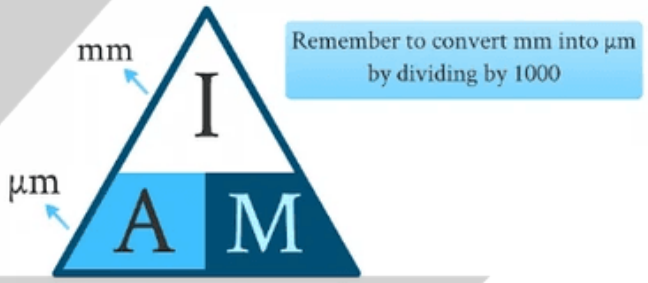
- Transmission Electron Microscope (TEM): 2D image, can see interior details
- Scanning Electron Microscope (SEM): 3D image, can only see surface structures

1.2. Magnification and Resolution

Magnification:

- the number of times larger an image is compared to the object's size.
- Depends on the power of the objective and eyepiece lens used.

Calculating magnification:



$$\text{Actual size} = \frac{\text{Image size}}{\text{Magnification}}$$

Eyepiece graticule:

- fitted into the eyepiece of the microscope and is used to measure objects.
- It has no units and is calibrated by the stage micrometer, which has an accurate scale (in mm) and provides reference dimensions.
- 1mm= 1000 µm
- 1µm= 1000 nm
- Use the same magnification when calibrating the eyepiece graticule and when using it to measure the specimen.

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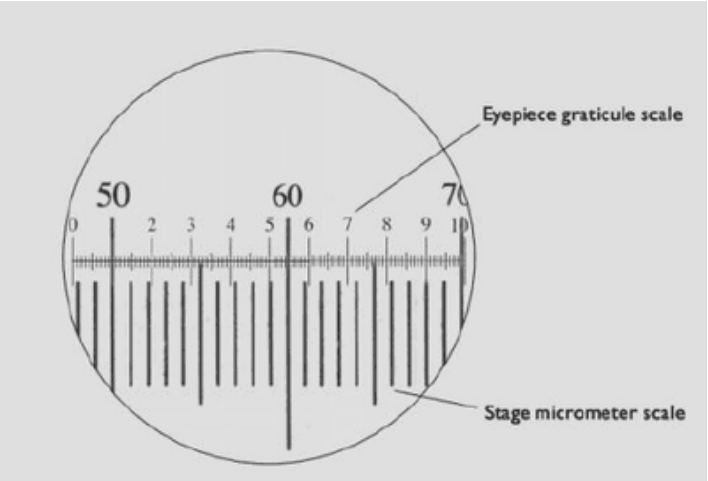
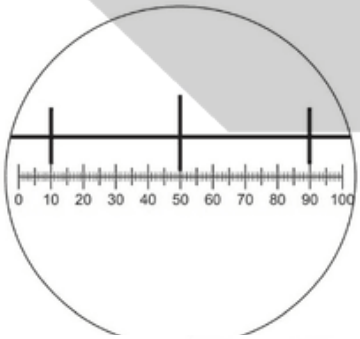
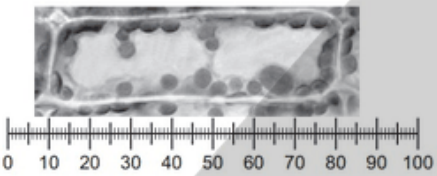


Calculation example:

3 The diagram shows a stage micrometer scale on which the small divisions are 0.1 mm. It is viewed through an eyepiece containing a graticule.



The stage micrometer scale is replaced by a slide of a plant cell.



What is the width of a chloroplast?

A 0.5 mm B 10 μm C 50 μm D 100 μm

- 0.1mm of SM = 40 div of EG
- 1 div of EG= 0.1/40
- 0.0025mm × 1000= 2.5μm
- 2.5μm×4 EGU of chloroplast width= 10μm

Resolution:

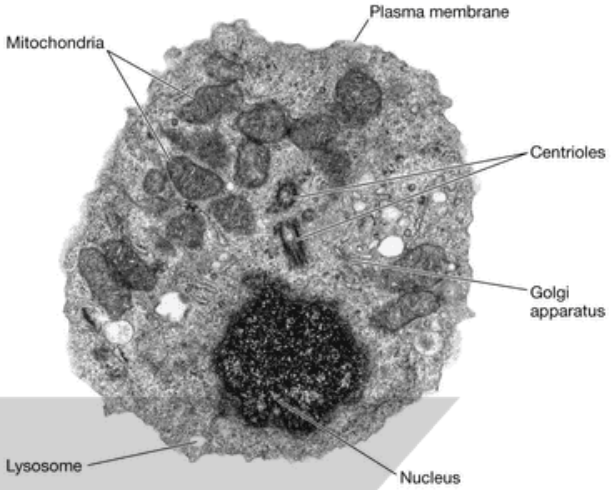
- The ability to distinguish between two separate points.
- The amount of detail that can be seen- higher resolution, higher detail.
- Note: a resolution of 10 will give a greater detail than a resolution of 2.

Limit of resolution:

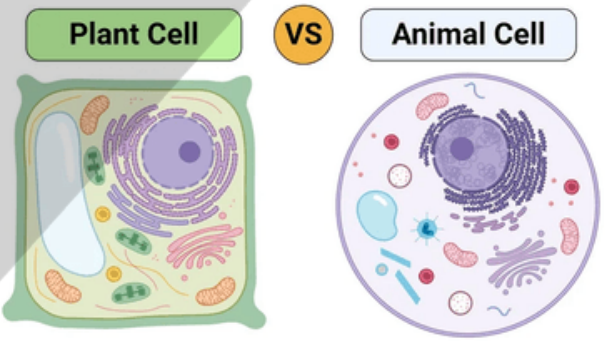
- Half the wavelength of radiation used to view the specimen.

1.3. Animal and Plant cells

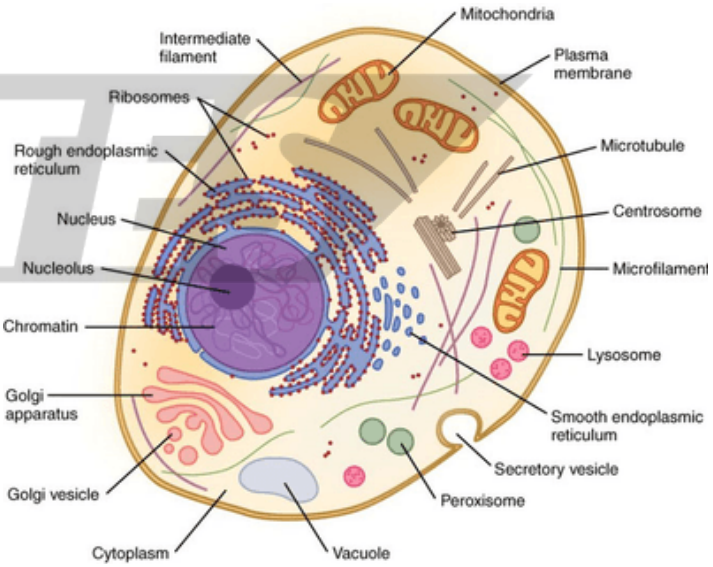
- **Ultrastructure:** the structure revealed by the electron micrograph.



- **Organelles:** functionally and structurally distinct parts of a cell, usually membrane-bound.



A generalised animal cell (20μm):



A generalised plant cell (40 μm):

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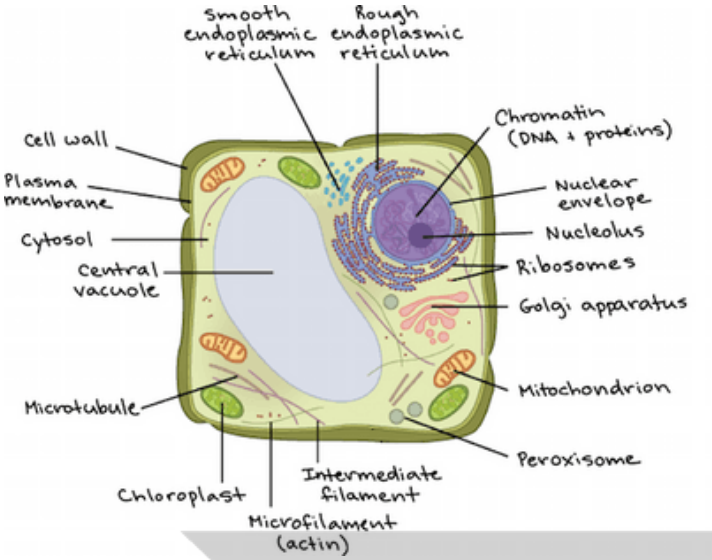
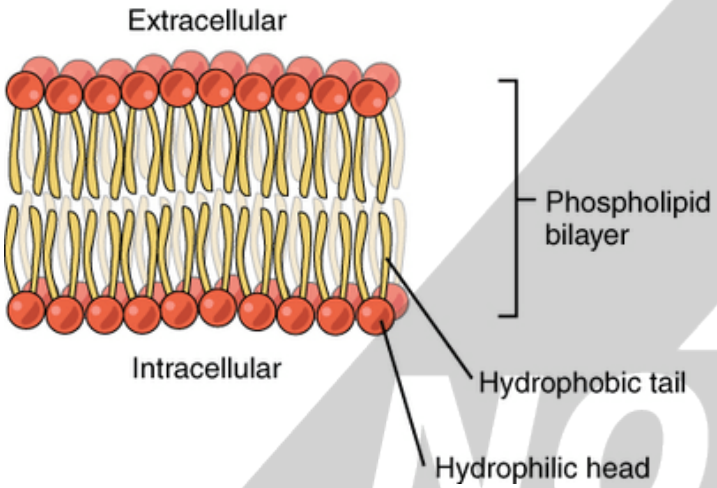


1.4. Cells as the Basic Unit of Living Organism

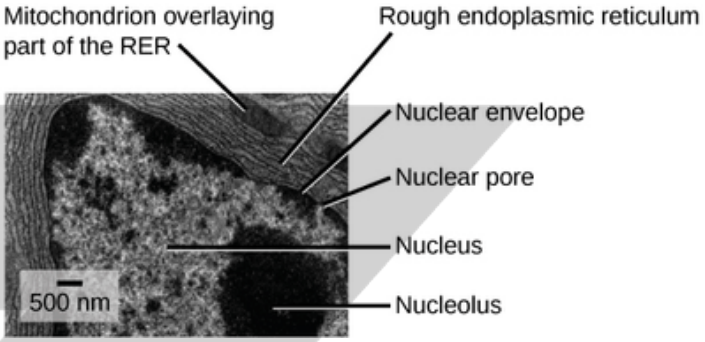
Cell Surface Membrane (7nm)

- Extremely thin with tri-laminar appearance
- It comprises phospholipid bilayers assembled with the hydrophilic phosphate heads facing the aqueous environment (inside and outside the cell) and the hydrophobic tails facing each other.



Function:

- The barrier between the cytoplasm and the external environment
- Cell signalling
- Cell recognition (surface antigens)
- Cell-to-cell adhesion
- Site for enzyme-catalysed reactions
- Anchoring the cytoskeleton
- Selection of substances that enter/leave the cell
- Formation of Hydrogen bonds with water for stability
- **Nucleus (10μm):** the largest organelle surrounded by the double-membraned nuclear envelope and is continuous with a rough endoplasmic reticulum.
- **Nuclear pore:** gaps in the nuclear envelope that allow exchange between the nucleus and cytoplasm.
  - **Substances leaving:** mRNA and ribosomes for protein synthesis.
  - **Substances entering:** protein to help make ribosome, nucleotide, ATP, & some hormones.
- **Chromosome:** contains the hereditary material DNA that is organised into genes which control the cell's activities and inheritance.
- **Nucleolus (0.2-0.5μm) one or more found (nucleoli) containing DNA and RNA, functioning to make ribosomes.**



- **Rough endoplasmic reticulum**: an extensive membrane system with 80S ribosomes are sites for protein synthesis, which produces the rough appearance. The R.E.R provides a pathway for the transport of materials through the cell.
  - Forms a complex 3D system of sheet-like membranes and tubes enclosing fluid-filled sacs.
  - Proteins made by ribosomes on RER enter sacs and move through them. Transport vesicles bud off from the RER and join, forming the Golgi body.
- **Smooth Endoplasmic reticulum:** site for lipid synthesis and steroids, e.g. cholesterol and reproductive hormones.
  - The meshwork of tubular membrane vesicles with fluid-filled sacs that have no ribosome on their surface

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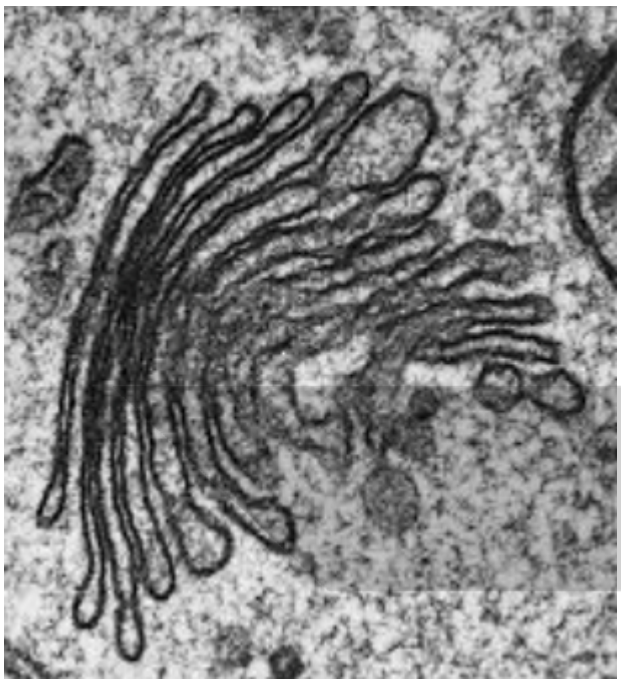
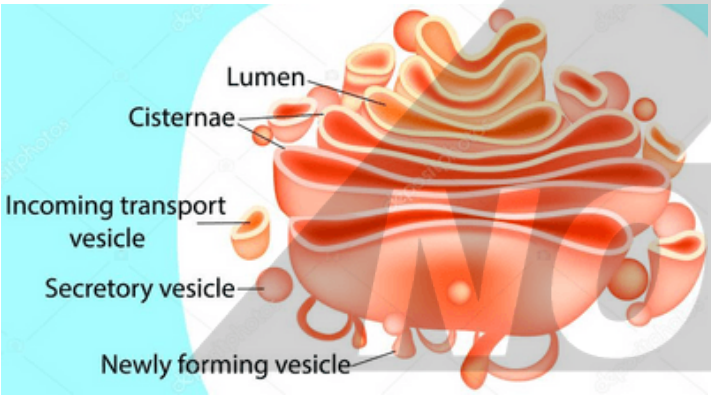
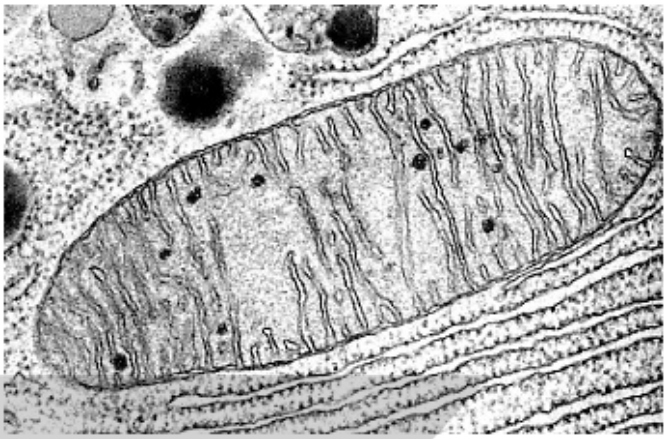
- **Golgi body/complex/apparatus:** a stack of flattened membranes enclosing hollow sacs (cisternae) formed by transport vesicles. Buds are off of the RER and broken down to form Golgi vesicles.



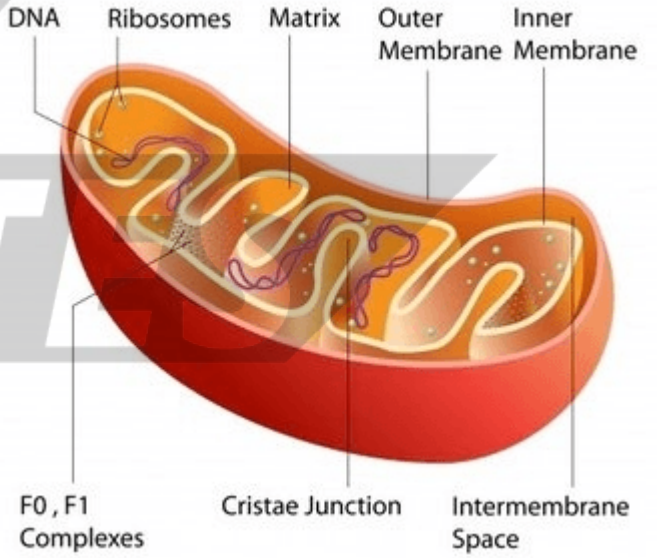
- Collects, processes, modifies and sorts molecules that are ready for transport in Golgi vesicles to other parts of the cell or out of the cell by:
  - **Secretion/exocytosis:** fusion of the vesicle with the plasma membrane to release content.
- Makes lysosomes, glycoproteins and functioning proteins.



- **Mitochondria (1μm):** surrounded by mitochondrial envelope; provides energy for aerobic respiration, synthesizes lipids, and more in areas requiring maximal energy.



- It has a matrix that contains 70S ribosomes and circular DNA, which is used to make some of the mitochondrion's proteins.
- **Cristae:** folding of inner membrane that projects into the interior solution, matrix.
- **Intermembrane space:** space between the two membranes.
- **Porin:** transports protein in the outer membrane and forms a wide aqueous channel, allowing water-soluble molecules from the cytoplasm to intermembrane space.
- **Inner membrane:** selective barrier controlling the entrance of ions and molecules into the matrix.



- **Role of adenosine triphosphate (ATP):**
  - Made up of 3 phosphate groups, a nitrogenous base and a ribose sugar.
  - The energy-carrying molecule produced in mitochondria spreads to parts where needed.
  - Energy is released by breaking ATP to ADP, a reversible hydrolysis reaction.

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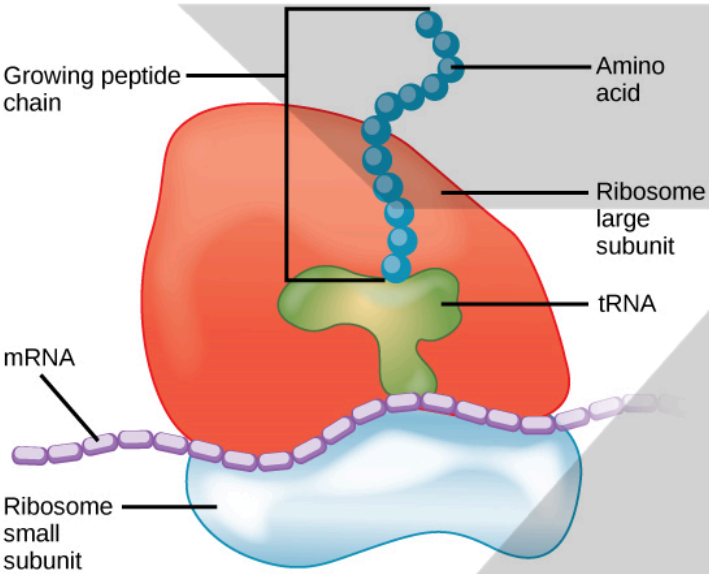
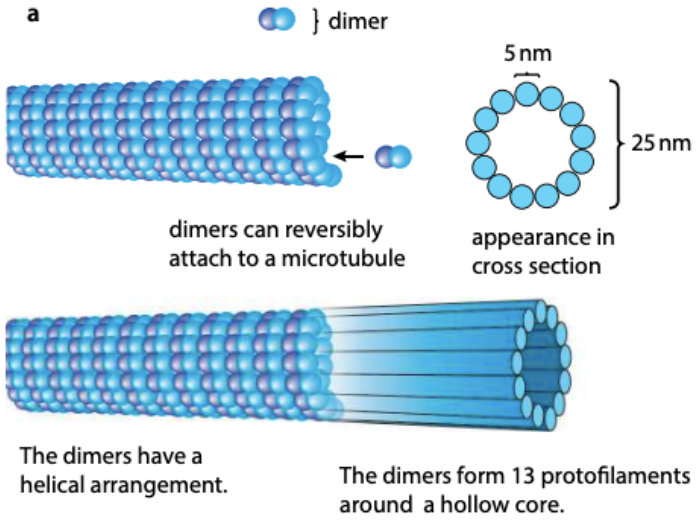
- **Endosymbiotic Theory:** Mitochondrion and chloroplast are bacteria that now live inside more giant cells of animals and plants, which is why chloroplast and mitochondrion have circular DNA.
- **Ribosomes:** the site at which mRNA (transcribed from the nucleus) is translated into polypeptides with the help of tRNA, therefore helping with protein synthesis. They are not membrane-bound and are made up of rRNA and proteins.
  - **80S ribosomes:** in the cytoplasm and R.E.R
  - **70S ribosomes:** in chloroplast and mitochondria.
  - **Small subunit:** site of translation (mRNA binds here and is read by this unit)
  - **Large subunit:** joins amino acids to form polypeptides.

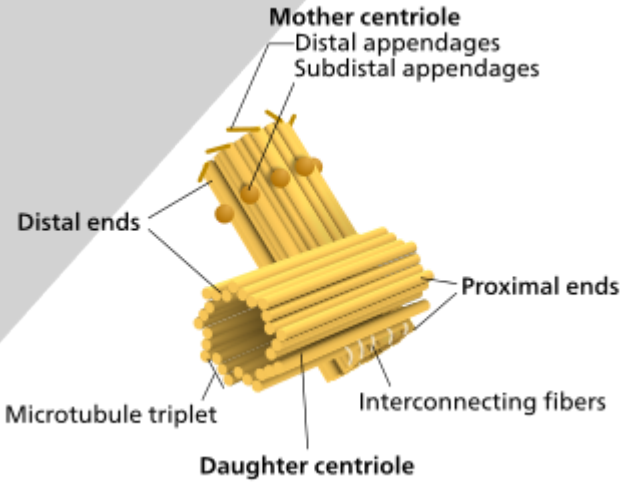


- **Lysosomes (0.1-0.5μm):** a single membrane with no internal structure in animal cells. They contain digestive (hydrolytic) enzymes kept separate from the rest of the cell to prevent damage.
  - Responsible for breaking down unwanted structures, e.g. old organelles or whole cells in WBC to digest bacteria.
- **Microtubules**
  - Are long hollow tubes that make up the cytoskeleton
  - Made up of alpha and beta tubulin that combine to form dimers.
  - Dimers join end to end to form protofilaments (polymerisation).
  - 13 protofilaments line up in a ring to form a cylinder with a hollow centre, i.e. microtubule.
  - It forms an intracellular transport system by moving along
  - Secretary vesicles, organelles, and cell components on its outer surface.
  - Allows movement of cilia in bronchi
  - Determines cell shape.



1.7. Organelles in Animal Cells

- **Centrosome:** a pair of centrioles at right angles that are involved in nuclear division and act as MTOCs.



- **Centriole:**
  - Formed by 9 triplets of microtubules
  - Microtubules extend from the centriole and attach themselves to the kinetochore of chromosomes, forming spindle fibres
  - centrioles duplicate, and a pair of centrioles then move to opposite poles of the cell (2 centrosome regions), thus separating sister chromatids during nuclear division.

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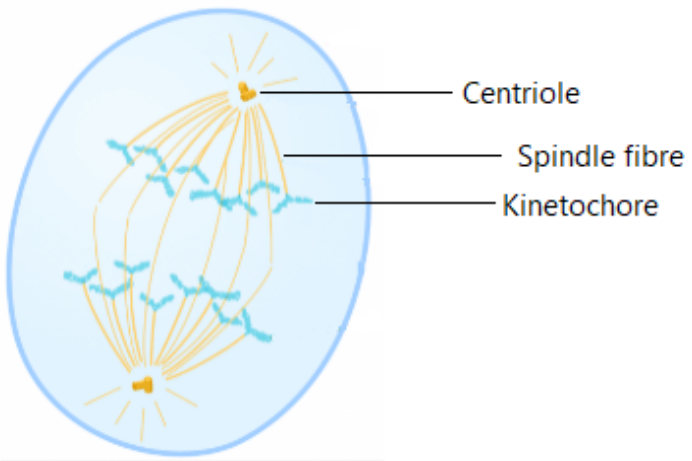
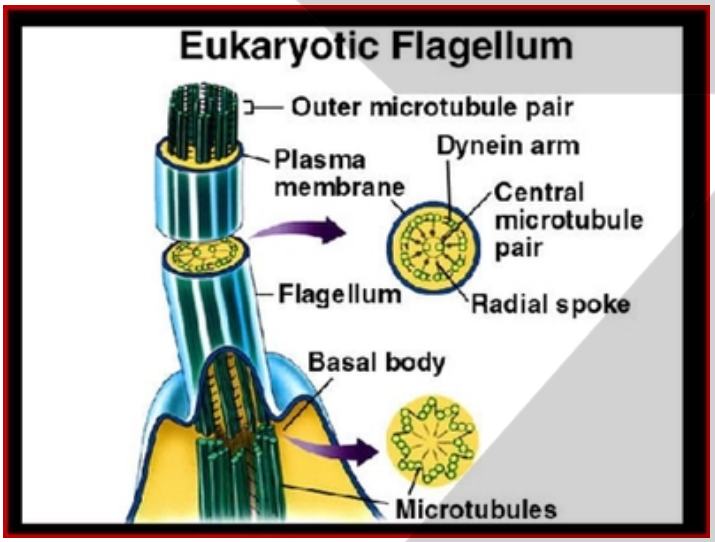


- Centrioles at the bases of cilia and flagella (basal bodies) act as MTOCs.
  - Microtubules extending from basal bodies into cilia and flagella help with their beating movements.



- Microvilli:
  - finger-like extensions
  - increase surface area for functions like reabsorption and absorption

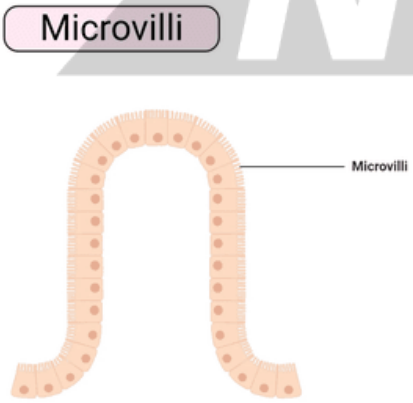
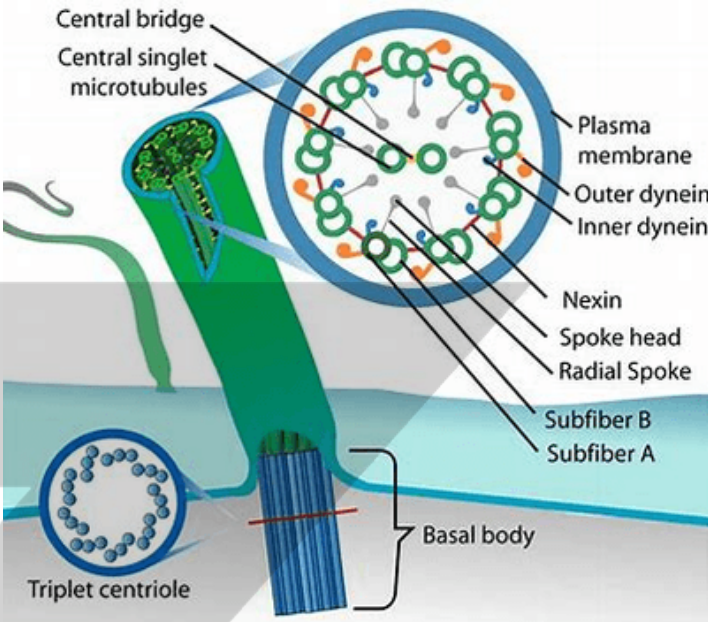
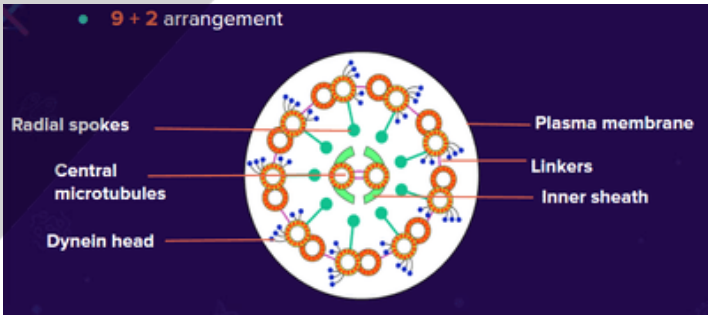


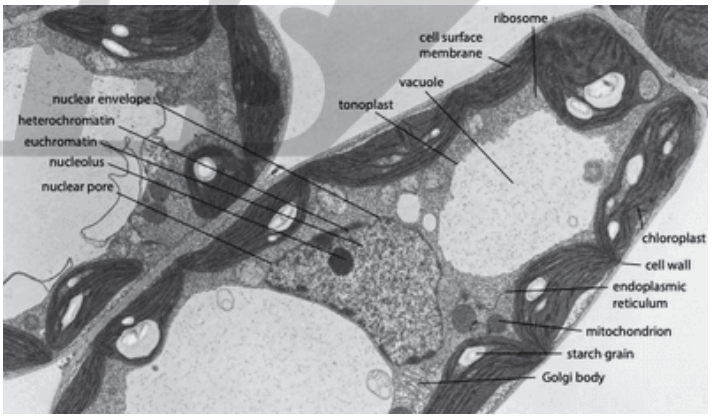
Figure: Microvilli, Image Copyright © Sagar Aryal, www.microbenotes.com

- Cilia:
  - Hair-like structure
  - Made of microtubules
  - Two central microtubules and a ring of nine microtubule doublets (MTD)
  - '9+2' structure





1.8. Organelles in Plant Cells



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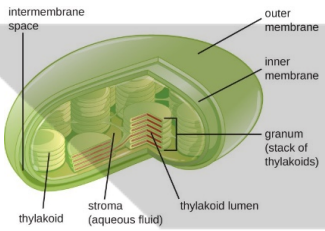
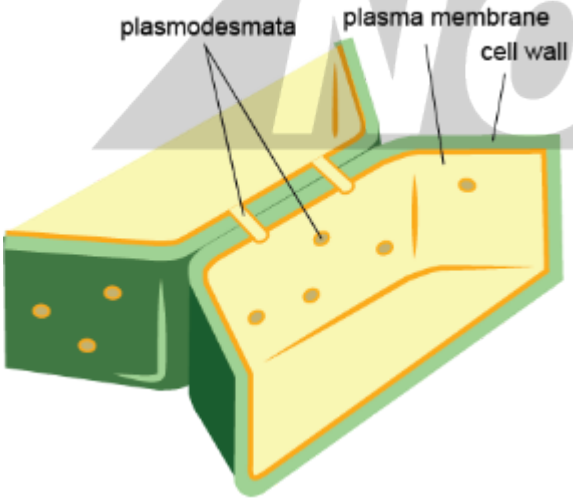
- **Chloroplasts (5-10μm):** This cell structure is only found in plant cells in the palisade mesophyll, spongy mesophyll, and surface of the stem, which carries out photosynthesis and ATP synthesis.
  - It has a double membrane and contains flattened sacs known as thylakoids.
  - Chlorophyll is embedded in thylakoid membranes.
  - Thylakoids are stacked on top of each other to form grana.
  - Grana are linked by lamella. These structures are present in a matrix called the stroma.
  - Contains starch grains, circular DNA, and 70S ribosomes.

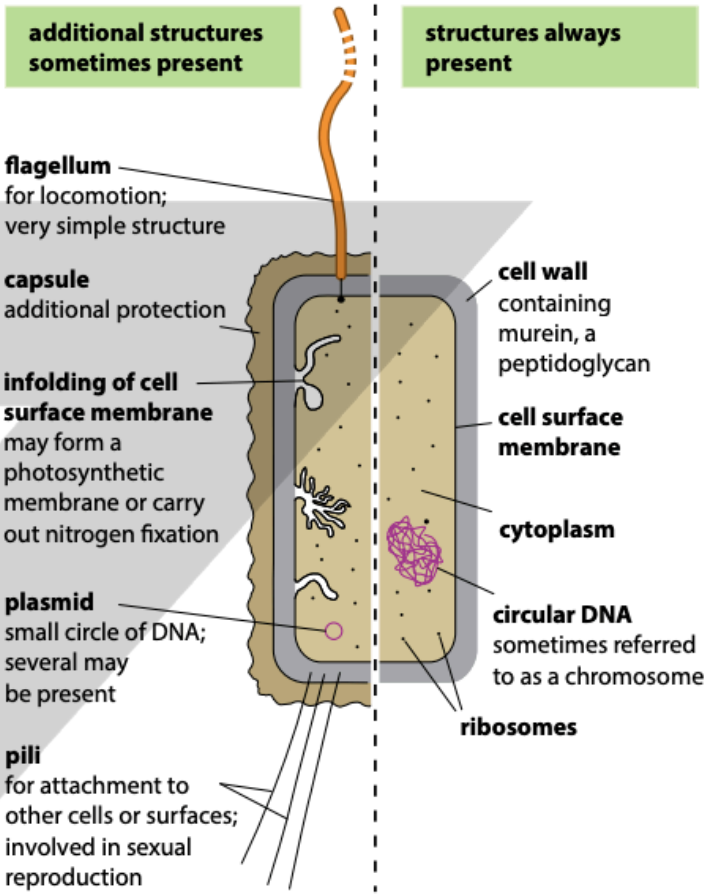


- **Cell wall (10 nm):** rigid as it contains cellulose fibers (polysaccharide).
  - It gives the cell its definite shape and prevents it from bursting (by osmosis), allowing turgidity.
  - It may be reinforced by lignin for extra strength.
  - Freely permeable.
- **Plasmodesmata:** pore-like structures found in cell walls that allow a link between neighboring cells by fine threads of cytoplasm.
- **Large vacuole and tonoplast:** surrounded by partially permeable tonoplast, has cell sap (fluid) that consists of enzymes, sugars, waste products, pigments, mineral salts, oxygen, and C02 and regulates osmotic properties.



1.9. Eukaryotes and Prokaryotes

- **Eukaryotes:** organisms with a true nucleus and have membrane-bound organelles e.g., animals, plants, fungi, and protoctists.
- **Prokaryotes:** organisms that lack a nucleus and have simpler structures e.g., bacteria.
- **Generalised bacterium:**



- **Comparing eukaryotes with prokaryotes:**

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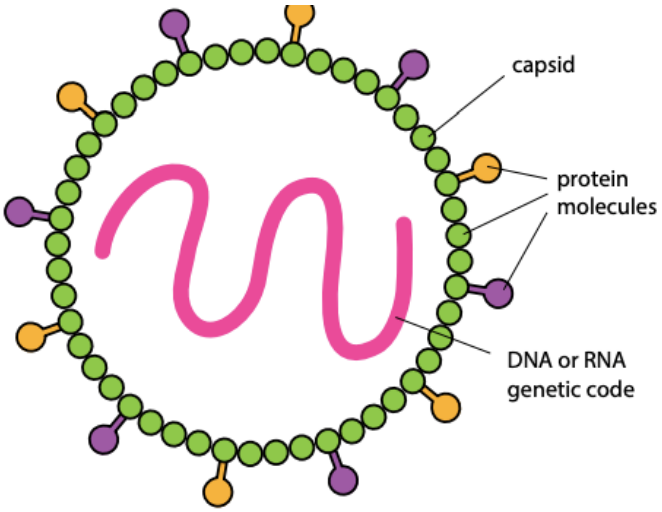
Feature	Prokaryote	Eukaryote
Average Diameter of Cell	0.5-5μm	10-100μm 1k-10k times volume of prokaryote
DNA	Circular Lies free in the cytoplasm Is naked	Linear Surrounded by nucleus Associated with histone, forming chromosome
Ribsome	70S (20nm)	80S (25nm)
ER	Absent	Present, to which ribosome may be attached
Organelles	Very few No membrane bound organelles unless formed by infoldings of plasma membrane	Many Single, double and no membrane bound organelles
Cell Wall	Murein, a peptidoglycan (polysaccharide with amino acid)	Cellulose and lignin in plants Chitin (nitrogen containing polysaccharide) in fungi
Cell division	Binary fission	Mitosis and cytokinesis
Transcription and translation	Simultaneously (as no nucleus)	Consecutively



1.10. Viruses

- 20-300nm
- non-cellular and parasitic as they reproduce by infecting and taking over living cells.
- The virus DNA/ RNA hijacks the protein synthesising machinery of the host cell, which then helps to make new viral proteins to make capsid.
- some have an outer phospholipid envelope

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